Passage — Product Requirements Document

Status	Draft v0.2 — for review by Ben Russell and Zach Demanche
Date	2026-10-21
Owners	@BenjaminSRussell (product), @zdemanche (engineering/security)
Method	Moonaan Scribe requirements workflow (understand → research → draft → validate → publish). Jira/Confluence research was unavailable this session; research covered public AX/SAP documentation and both owners' GitHub accounts (no prior ERP work found).
Companion	roadmap/waves.json is the machine-readable source for the 50 waves; roadmap/README.md is generated from it.
Thesis	Emulate the top-tier manufacturing ERPs (S/4HANA, D365 F&SCM, Infor LN, Epicor Kinetic, IFS, Plex) feature-for-feature, then be **10× easier, faster, more accurate and more open** — see Part 0A.

Document map

This master document (Part 0) holds the problem, goals, scope, delivery model, phase-level requirements and NFRs. The detailed parts hold wave-level requirements and are assembled into `docs/PRD-full.md` by `scripts/build_prd.py`.

Part	File	Content
0	`docs/PRD.md` (this file)	Problem, goals, personas, principles, scope map, delivery model, phase-level FRs, NFRs, risks
0A	prd/part-00-thesis.md	The 10× thesis: pain-point ledger (usability, speed, accuracy, openness), interoperability contract, what we keep from AX and SAP
1	prd/phase-0-foundation.md	Waves 1–5: pipeline, identity/tenancy, metadata kernel, events/workflow, shell/audit/observability
2	prd/phase-1-master-data.md	Waves 6–10: enterprise structure, party model, products, engineering data, migration framework
3	prd/phase-2-financial-core.md	Waves 11–15: universal journal, currencies/ledgers, tax, close, reporting
4	prd/phase-3-procure-to-pay.md	Waves 16–20: inventory ledger, purchasing, receiving/matching, AP/payments, costing/supplier portal
5	prd/phase-4-manufacturing.md	Waves 21–25: production orders, MRP, scheduling, process/lean, shop floor/IoT
6	prd/phase-5-order-to-cash.md	Waves 26–30: pricing, orders/ATP, fulfillment, billing/rev rec, AR/collections — go-live candidate
7	prd/phase-6-logistics.md	Waves 31–35: WMS, mobile, quality, transportation, EDI/e-invoicing/integration hub
8	prd/phase-7-treasury-controlling.md	Waves 36–40: cash/bank, fixed assets, budgeting/cost accounting, product costing, intercompany/consolidation
9	prd/phase-8-projects-people-service.md	Waves 41–45: projects, HR, payroll/expenses, EAM, field service
10	prd/phase-9-intelligence-golive.md	Waves 46–50: data platform, AI agents, localization/compliance, migration factory, scale/go-live
11	prd/appendices.md	Glossary, cross-module data model, event/API conventions, security model, NFR detail, numbering, localization, testing standards, traceability, references, competitive benchmark

Part	File	Content
12	prd/coverage-ax.md	Completeness oracle: every AX 2012 R3 / D365 F&SCM module → forms, inquiries/reports, periodic jobs, setup forms, each mapped to a wave or flagged GAP; gap register
13	prd/coverage-sap.md	Completeness oracle: every SAP ECC / S/4 component → transactions, reports, periodic processing, IMG customizing, each mapped to a wave or flagged GAP; gap register

Each wave section follows the fixed template in prd/TEMPLATE.md: scope and pain points solved → AX/SAP parity map → functional requirements → data model → APIs/events → UX → business rules → acceptance criteria → test plan → deployment → migration → dependencies/risks/open questions → parameters and setup → inquiries and reports → printed/electronic documents → periodic jobs → duties and privileges → message catalog → performance budgets → localization hooks.

Completeness rule. A wave is not specified until every row of Parts 12 and 13 that maps to it is either covered by a numbered FR in that wave's section or explicitly marked SUPERSEDED/OUT-OF-SCOPE with a reason. The gap registers (12.99, 13.99) must be empty before Phase 0 starts.

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1. Problem statement

Microsoft Dynamics AX (Axapta → AX 2012 → D365 Finance & Supply Chain) and SAP (ECC → S/4HANA) are the systems of record for most mid-to-large manufacturers, distributors and services firms. Both are:

- **Closed and expensive to change** — customization lives in proprietary languages (X++, ABAP) and licensing/upgrade costs dominate TCO.
- **Architecturally dated** — monolithic application servers, synchronous batch frameworks, and vendor-locked cloud operation (LCS, RISE).
- **Hard to integrate and extend with AI** — bolted-on copilots that can't act safely across the system.

Passage is a ground-up rebuild of the complete feature surface of AX/SAP as an open-architecture, cloud-agnostic platform: Kubernetes-native, Postgres-per-domain, Kafka-eventing, metadata-driven, with AI agents as first-class, governed actors.

2. Goals and non-goals

Goals

1. **Feature parity**: every AX 2012 / D365 F&SCM and SAP ECC / S/4 module in scope (§5) has an equivalent in Passage, mapped wave-by-wave.
2. **Provable delivery**: every one of the 50 waves ends with an artifact that is (a) **provable** — automated tests and fixtures that demonstrate correctness, (b) **testable** — a stakeholder can exercise it in a running environment, and (c) **deployable** — it ships to a Kubernetes environment through the GitOps pipeline.
3. **Sequential, top-down**: waves are strictly ordered; each wave depends only on earlier waves (dependency graph is a chain with stated back-references). A manufacturer can go live at the end of wave 30; full AX/SAP parity at wave 50.
4. **Cloud-agnostic and self-hostable**: runs on any conformant Kubernetes (EKS/GKE/AKS/bare-metal/k3d) with no provider-specific managed services in the critical path.
5. **Extensible without forking**: customers extend entities, workflows, forms and reports through metadata and sanctioned extension points, never by overlayering code.

6. **Financially rigorous**: a single append-only universal journal is the only source of truth; every subledger reconciles to it by automated check.
7. **10× on four measured axes** (defined in Part 0A): easier (≤ 1/10 the clicks per task; new user productive in < 30 min), faster (MRP incremental < 2 min, continuous close, real-time statements), more accurate (zero unexplained subledger-to-GL breaks, < 0.5% duplicate master data), open (any system on the customer's network connects in < 1 day through REST/GraphQL/Kafka/webhooks/MCP/EDI/OPC-UA/SFTP with one identity and one policy engine).
8. **Best-of-breed manufacturing**: planning, scheduling, shop floor, quality, costing and warehouse are benchmarked against S/4HANA, D365 SCM, Infor LN, Epicor Kinetic, IFS Cloud and Plex (Appendix L), not only against AX 2012.

Non-goals (v1)

- Replacing CRM/marketing automation suites (we integrate with Salesforce/HubSpot; a native CRM is v2).
- Retail POS and e-commerce storefront (we integrate with Shopify/Medusa; native POS is v2).
- Country packs beyond the first set (US, UK, DE, FR, IN, MX, CA) — framework ships in wave 48; more packs are v1.x.
- Supporting SAP's or Microsoft's proprietary extension languages. Migration tooling moves **data**, not X++/ABAP code.

3. Users and personas

Persona	Primary jobs	Waves that serve them first
Platform engineer / SRE	Deploy, observe, upgrade, scale, secure	1–5, 48, 50
System administrator	Tenants, users, roles, extensions, number sequences, audit	2, 3, 5, 48
Master-data steward	Parties, products, BOMs, org structure, migration	6–10
Controller / accountant	Journal, close, statements, tax, FX, subledgers	11–15, 19, 30, 36–40
Buyer / AP clerk	Requisition to payment	17–20
Production planner / supervisor	Orders, MRP, scheduling, shop floor	21–25
Sales rep / order desk / AR clerk	Quote to cash	26–30
Warehouse manager / operator	WMS, mobile, shipping, quality, transport	28, 31–35
Project manager / HR / payroll admin	Projects, people, payroll, assets, service	41–45
Analyst / executive	Dashboards, KPIs, copilots	46–47
Implementation partner	Migration, localization, cutover	10, 48–50

4. Product principles

1. **Journal first.** Nothing is financially true until it is in the universal journal. Subledgers are projections; they reconcile automatically every day.
2. **Metadata over code.** Entities, forms, workflows, reports and extensions are data. Code generators produce schema, API, clients and UI from it.
3. **Events are the integration surface.** Every state change is an immutable CloudEvent on Kafka via transactional outbox — no dual writes, no polling.

4. **Everything is tenant-scoped and policy-authorized.** Postgres RLS + OPA on every request; agents inherit (never exceed) their user's permissions.
5. **Idempotent, replayable, auditable.** Posting APIs take idempotency keys; audit is derived from CDC, not from application code that can forget.
6. **Ship the thin vertical slice.** A wave is done when a stakeholder can run the business process on a deployed environment with reconciled numbers — not when the code is written.

5. Scope map — AX/SAP modules → Passage phases

AX / D365 module	SAP module	Passage phase (waves)
AOT, security, workflow, batch, LCS, number sequences, DB log	DDIC, PFCG, Business Workflow, SM36, STMS, SNRO, CDHDR	**0 Foundation (1–5)**
Org model, global address book, products, BOM/route, DIXF	Enterprise structure, Business Partner, material master, CS/CA, LSMW	**1 Master data (6–10)**
General ledger, dimensions, currencies, tax, period close, financial reporting	FI-GL/ACDOCA, ledgers, FTXP, OB52, FSV	**2 Financial core (11–15)**
Inventory management, procurement & sourcing, AP, inventory costing	MM-IM, MM-PUR, LIV, FI-AP, F110, material ledger	**3 Inventory & P2P (16–20)**
Production control, master planning, scheduling, process/lean, MES	PP, PP-PI, REM, KANBAN, PP/DS, MII	**4 Manufacturing & planning (21–25)**
Trade agreements, sales, shipping, invoicing, rev rec, AR, collections	SD pricing, SD orders, LE-SHP, billing, RAR, FI-AR, FSCM	**5 Order-to-cash (26–30)**
Warehouse management, mobile app, quality, TMS, AIF/ER/EDI	EWM, RF, QM, TM, IDoc/PI, DRC	**6 Warehouse, quality & logistics (31–35)**
Cash & bank, fixed assets, budgeting/cost accounting, costing sheets, intercompany/consolidation	Cash mgmt, FI-AA, CO, product costing, Group Reporting	**7 Treasury, assets & controlling (36–40)**
Projects, HR, payroll, expense, asset management, service	PS, HCM/SF, PY, Concur, PM, CS/FSM	**8 Projects, people & service (41–45)**
Entity store/Power BI, Copilot, localizations/SoD, upgrade tooling, LCS prod	BW/SAC, Joule, country versions/GRC, conversion tools, Basis	**9 Intelligence, compliance & go-live (46–50)**

Full per-wave mapping with references is in roadmap/waves.json (`replaces` field).

6. Delivery model — what a "wave" is

A wave is a 4–8 week increment with four mandatory outputs, enforced by the wave issue template and the Definition of Done:

Output	Definition	Evidence required to close the wave
Deliverable	The capabilities listed in the wave's `deliverable` array, built on the agreed stack.	Merged PRs linked from the wave issue.
Proof	The automated tests, fixtures, property tests and performance targets in `proof`.	CI run link; fixture datasets committed; perf report.
Deploy	What is running where, as stated in `deploy`.	Argo CD app healthy in `stage`; demo URL; recorded walkthrough for phase-exit waves.
References	The AX/SAP documentation and open-source/standards we built against.	Linked in the wave issue; ADRs for any deviation.

Sequencing rules

- Wave *n* may start when wave *n−1* has passed Proof and Deploy; overlapping *build* is allowed, overlapping *close* is not.
- Every 5th wave is a **phase-exit wave** with a stakeholder sign-off (demo + reconciliation evidence).
- Wave 30 is the **first go-live candidate** (manufacturer buy → make → sell → collect). Wave 50 is **production go-live + v1.0**.

7. Functional requirements (phase level)

Phase-level requirements below are the contract; wave-level detail (deliverables, proofs, references) is normative in `roadmap/waves.json` and rendered in `roadmap/README.md`.

FR-0 Foundation (waves 1–5)

- **FR-0.1** The platform deploys to any conformant Kubernetes via GitOps with signed images, SBOMs and digest-pinned rollback.
- **FR-0.2** Identity is federated (OIDC/SAML/SCIM); authorization is AX-style roles/duties/privileges compiled to OPA policies; tenant isolation is enforced by Postgres RLS.
- **FR-0.3** Business entities are declared in metadata that generates schema, API, clients, validation and UI; customers extend entities without code.
- **FR-0.4** All state changes publish CloudEvents via transactional outbox; approvals and batch jobs run on a durable workflow engine with timers, retries and human tasks.
- **FR-0.5** A single web shell renders any entity as accessible (WCAG 2.1 AA) list/detail/grid; number sequences support gap-free mode; audit is field-level and immutable; every request is traceable end-to-end.

FR-1 Master data (waves 6–10)

- **FR-1.1** Legal entities, operating units, sites, warehouses and effective-dated multi-purpose hierarchies; fiscal calendars; ISO reference data; UoM conversions.
- **FR-1.2** One party per real-world organization/person with layered roles, duplicate detection and merge, tax-id validation, governance workflow.
- **FR-1.3** Products with product/storage/tracking dimension groups, released products per entity, attributes/classification, GS1 identifiers, configurator.
- **FR-1.4** Multi-level versioned BOMs and formulas, routings/operations/resources/calendars, engineering change management with where-used.
- **FR-1.5** A migration framework with AX and SAP connectors, declarative mapping, validation, reconciliation and idempotent re-runs.

FR-2 Financial core (waves 11–15)

- **FR-2.1** Append-only double-entry universal journal with dimensions, account structures, idempotent atomic posting, balance projections rebuilt from the journal.
- **FR-2.2** Three-currency amounts per line, rate types/providers, parallel ledgers, FX revaluation and translation.
- **FR-2.3** Tax determination/calculation/posting for sales & use, VAT/GST and withholding; pluggable external providers; settlement and statutory extracts (SAF-T).
- **FR-2.4** Period control per ledger × module, close checklists, allocations, accrual schemes, year-end close.

- **FR-2.5** Statement designer and standard statements with drill-down to source document; audit trail reporting.

FR-3 Inventory & P2P (waves 16–20)

- **FR-3.1** Immutable inventory transactions with physical/financial status and dimensions; journals; reservations; cycle counting; GL integration.
- **FR-3.2** Requisition → RFQ → agreement → PO with policy-driven approvals, versioned change management and vendor pricing.
- **FR-3.3** Receiving with GR/IR accrual; vendor invoice capture; 2/3-way matching with tolerances and exception workflow; ERS.
- **FR-3.4** AP open-item settlement, payment proposals, ISO 20022/NACHA/check payment media, positive pay, prepayments.
- **FR-3.5** FIFO/LIFO/weighted-average/standard costing with inventory close and GL reconciliation; supplier portal.

FR-4 Manufacturing & planning (waves 21–25)

- **FR-4.1** Production order lifecycle with picking, route consumption, RAF, WIP and variance settlement; subcontracting.
- **FR-4.2** MRP/MPS with coverage groups, planned orders, action/delay messages, pegging, statistical forecasting.
- **FR-4.3** Finite/infinite capacity scheduling with Gantt and optional CP-SAT optimization.
- **FR-4.4** Process manufacturing (formulas, potency, co/by-products, catch weight, batch attributes) and lean (production flows, kanban).
- **FR-4.5** Shop-floor kiosk with time registration, IoT ingestion (OPC-UA/MQTT), OEE, offline tolerance.

FR-5 Order-to-cash (waves 26–30)

- **FR-5.1** Deterministic pricing engine with condition tables/access sequences, discounts, promotions, rebates, explain.
- **FR-5.2** Quotations, sales orders, ATP/CTP, credit management, holds, multi-channel intake.
- **FR-5.3** Release, pick/pack/ship, packing slips with COGS, carrier integration, drop-ship.
- **FR-5.4** Invoicing (one-off, consolidated, subscription), ASC 606/IFRS 15 revenue recognition.
- **FR-5.5** Cash application, collections/dunning/disputes, RMA, rebate settlement, customer portal with payments. **Go-live candidate.**

FR-6 Warehouse, quality & logistics (waves 31–35)

- **FR-6.1** Location directives, work templates, waves, license plates, replenishment, load planning.
- **FR-6.2** Mobile scanner app with configurable flows, GS1 parsing, offline queue, label printing.
- **FR-6.3** Quality orders by rule, sampling, usage decisions, inventory blocking, NCR/CAPA, CoA, supplier scorecards.
- **FR-6.4** Load/route planning, rating, tendering, tracking, freight reconciliation, landed cost.
- **FR-6.5** EDI (X12/EDIFACT), Peppol e-invoicing, integration hub (APIs, webhooks, catalog, developer portal).

FR-7 Treasury, assets & controlling (waves 36–40)

- **FR-7.1** Bank accounts, statement import and auto-reconciliation, cash positioning/forecasting, bank connectivity with vaulted keys.

- **FR-7.2** Fixed assets with multi-book depreciation, full transaction set, CIP from projects, IFRS 16 leases.
- **FR-7.3** Budgeting with workflow and budget control on commitments; cost accounting ledger with allocations and ABC.
- **FR-7.4** Costing sheets, BOM cost roll-up, standard cost activation and revaluation, variance decomposition.
- **FR-7.5** Intercompany trade/accounting with mirror documents and reconciliation; group consolidation with eliminations and translation. **Finance parity.**

FR-8 Projects, people & service (waves 41–45)

- **FR-8.1** WBS, resources, hour/expense/item journals, T&M/fixed-price/milestone billing, WIP and POC accounting, EVM.
- **FR-8.2** Workers/positions/jobs with effective dating, absence, benefits, self-service, talent basics, PII controls.
- **FR-8.3** Country-pluggable gross-to-net payroll (US + one EU reference) and expense management with policies and card feeds.
- **FR-8.4** Functional locations/assets, preventive and condition-based plans, work orders with costs to asset/GL, IoT triggers.
- **FR-8.5** Installed base, agreements/SLAs, cases, field work orders, dispatch optimization, technician mobile app.

FR-9 Intelligence, compliance & go-live (waves 46–50)

- **FR-9.1** CDC lakehouse (Iceberg) + semantic layer with governed metrics, embedded dashboards, document designer, catalog/lineage.
- **FR-9.2** MCP-exposed APIs; copilot UI; governed agents with approval gates, evals in CI, cost budgets, model-agnostic provider layer.
- **FR-9.3** Localization framework and first country packs; electronic reporting designer; SoD engine; SOC 2 control evidence; privacy requests.
- **FR-9.4** Transactional migration packs for AX and SAP, parallel-run diff harness, cutover runbooks, coexistence adapters.
- **FR-9.5** Scale/perf targets met, multi-region DR, operations and documentation complete, first customer in production, v1.0 released.

8. Non-functional requirements

Area	Requirement
Availability	99.9% per tenant (v1), 99.95% target with multi-region (wave 50). RPO 5 min, RTO 1 h.
Performance	Journal posting 5k lines/s sustained; generated entity reads p95 < 50 ms; pricing p95 < 20 ms; order intake p99 < 500 ms; dashboard lag p95 < 60 s.
Scale	10k concurrent users; 1M journal lines/h; 100k order lines/h; 50k-item MRP in < 10 min.
Security	OIDC/SAML/SCIM; OPA authorization on every request; Postgres RLS; signed images/SBOM; secrets in Vault; SoD engine; pen-test clean at high/critical before wave 48 exit.
Compliance	SOC 2 TSC control mapping, GDPR/CCPA subject requests, SAF-T, Peppol, ISO 20022, GS1.
Accessibility	WCAG 2.1 AA for every generated form and every hand-built workspace (axe in CI).
Observability	OpenTelemetry traces/metrics/logs for 100% of services; SLO dashboards; alerting.
Portability	No provider-managed service in the critical path; Helm + Terraform/OpenTofu modules per provider; k3d local parity.

Area	Requirement
Extensibility	Entity/enum/field/workflow/form/report extensions as metadata; versioned public APIs with additive-change guarantees.
Data integrity	Every subledger reconciles to the universal journal daily by automated check; no negative inventory unless policy allows; gap-free fiscal sequences.
Auditability	Field-level immutable audit from CDC; every agent action logged with inputs/outputs/approver.

9. Technical architecture (summary — see ADR-0001)

- **Runtime**: Kubernetes (any conformant distribution), Argo CD GitOps, Envoy Gateway, cert-manager.
- **Services**: Go for ledgers/engines (journal, inventory, pricing, MRP, scheduling), TypeScript (Node) for workflow/integration/UI-adjacent services, Python for forecasting/ML/agent evals.
- **Data**: Postgres per domain (CloudNativePG), RLS, Atlas migrations generated from metadata; Apache Iceberg lakehouse with Trino/ClickHouse for analytics.
- **Eventing**: Kafka (Strimzi), Debezium CDC + transactional outbox, CloudEvents + AsyncAPI, schema registry.
- **Workflow**: Temporal for approvals, batch, agents.
- **Identity/authz**: Keycloak, OPA/Rego.
- **Frontend**: React + module federation shell; PWA for shop floor, warehouse and field mobile.
- **AI**: MCP server over Passage APIs; Claude as default model via a provider-agnostic layer; eval harness in CI.
- **Observability**: OpenTelemetry → Prometheus/Tempo/Loki/Grafana.

10. Success metrics

Milestone	Metric
Wave 5	Platform demo: deploy → login → extend entity → approve workflow → trace a request, all live.
Wave 15	A period closed in Passage with statements that tie to the trial balance to the cent.
Wave 30	Demo manufacturer runs one simulated month buy → make → sell → collect; all subledgers reconcile; stakeholder sign-off.
Wave 40	Group consolidation produced; finance parity sign-off by a practicing controller.
Wave 50	First customer month-end in production; load/DR targets met; v1.0 released.
Ongoing	Every wave: CI green, proofs committed, Argo app healthy in stage, references linked.

11. Risks and mitigations

Risk	Mitigation
Scope is enormous (50 waves)	Strict sequencing; each wave independently demoable; phase-exit sign-offs; `waves.json` is the single plan of record and is re-estimated each phase exit.
Accounting correctness	Fixture sets reviewed by a practicing accountant per finance wave; daily automated subledger-to-GL reconciliation from wave 16 onward.
Postgres-per-domain vs cross-domain transactions	Sagas over Temporal + outbox; the journal posting API is the only cross-domain financial truth; no distributed 2PC.
Cloud-agnostic ambition increases ops burden	Reference stack is entirely CNCF/open-source with Helm charts; provider-specific Terraform modules are thin and optional.

Risk	Mitigation
Payroll/tax/localization legal exposure	Ship provider adapters (Avalara/ADP-style) alongside internal engines; country packs gated by regulator-schema fixtures.
AI agent misuse	Agents inherit user permissions via OPA, approval gates, evals in CI, kill switch, cost budgets.
Two-person team	GitHub Projects + wave issues as the operating system; AI-assisted generation (Moonaan sdlc `generate-architecture` → `generate-backlog` → `generate-code`) per wave.

12. Open questions

1. Licensing model for v1.0 (open-core vs. source-available vs. commercial) — decide before wave 35 (integration hub/public APIs).
2. First real customer: identify by wave 20 to shape go-live scope at wave 30.
3. Which EU country is the payroll/localization reference (DE vs. UK)?
4. Hosting of the demo environments (which provider first, even though the stack is agnostic)?

13. Next steps in the Moonaan SDLC pipeline

1. **This PRD** is the requirements source.
2. Run `/sdlc:generate-architecture docs/PRD.md docs/architecture/` per phase to produce the architecture document and compiled fact sheets.
3. Run `/sdlc:generate-backlog` per wave to turn `waves.json` entries into epics/stories with dependency ordering (GitHub Issues are the backlog store for now; Jira when the Moonaan proxy is reachable).
4. Run `/sdlc:generate-code` per story.

Validation checklist (Scribe step 4)

- [x] Every functional requirement is traceable to a wave with a provable, testable, deployable output.
- [x] Every wave names the AX and SAP features it replaces (grounding).
- [x] Every wave links external references (standards, vendor docs, open-source implementations).
- [x] Non-functional requirements are measurable.
- [x] Sequencing and go-live milestones are explicit.
- [ ] Reviewed by @zdemanche.
- [ ] Accountant review of finance-wave fixtures (owner TBD).

Part 0A — The 10× Thesis: What Top-Tier Manufacturing ERPs Get Wrong and How Passage Fixes It

Passage is not a cheaper AX or a friendlier SAP. It emulates the full capability surface of the best manufacturing ERPs — SAP S/4HANA, Dynamics 365 Finance & Supply Chain, Infor LN / CloudSuite Industrial, Epicor Kinetic, IFS Cloud and Plex — and then removes the structural reasons those systems are slow, error-prone, expensive to use and closed. "10×" is measured, per wave, on four axes:

Axis	What we measure	Baseline (typical AX 2012 / D365 / S/4 deployment)	Passage target
Easier	Clicks and screens per task; training hours to competence; share of tasks completed without a super-user	Sales order with 5 lines: 40–60 clicks across 4 forms; AP clerk training 3–5 days	≤ 1/10 of clicks; a new user completes a guided task in < 30 min
Faster	Latency of the business answer (not the screen): MRP, inventory close, ATP, period close, report	MRP regeneration 2–8 h nightly; inventory close 1–6 h; month-end close 5–10 business days	MRP incremental < 2 min, full < 10 min; close continuous; statements real-time
More accurate	Reconciliation breaks per month; silent rounding/FX errors; duplicate master data; manual journal share	Subledger-to-GL breaks every close; 3–8% duplicate customers/vendors; 20–40% of journals are corrections	Zero unexplained breaks (checked daily by machine); < 0.5% duplicates; corrections < 5%
Open	Days to connect a new system; share of data reachable by API/event; integration projects per year	Weeks per interface (AIF/IDoc mapping projects); large share of logic reachable only from the UI	Any system on the network connects in < 1 day with no middleware; 100% of entities and events reachable

0A.1 The structural pain-point ledger

Each pain point below names the AX/SAP mechanism that causes it, the root cause, the Passage design that removes it, and the wave where it ships. Wave files repeat this at wave granularity under "Pain points solved (10× lens)".

A. Usability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
U1	Dozens of forms to complete one business task; fields scattered across tabs and FastTabs	AX `SalesTable` form with 20+ tabs; SAP `VA01` header/item/schedule-line screens	Forms mirror tables, not tasks	Task-oriented screens generated from metadata with the *task* as the unit (order intake, receipt, close task); line grids with inline edit; the full document on one scrollable page	5, 27
U2	Lookups require knowing the code (vendor account, main account, item number)	`MainAccount` lookup in `LedgerJournalTrans`; SAP GL account search help	Keys are the primary identifier; search is secondary	Global typeahead on name/description/any attribute, ranked by recency and role; keys shown but never typed	5

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
U3	Personalization is per-user, fragile, lost on upgrade	AX usage data (`SysLastValue`); SAP GUI variants	Personalization stored outside metadata	Saved views are first-class, versioned, shareable, role-defaulted; survive upgrades because they target metadata ids	5
U4	Batch jobs hide status; users poll "Batch jobs" form to see if posting finished	`BatchJob`/`BatchJobHistory`; SAP `SM37`	Fire-and-forget batch framework with no UI contract	Every long-running action returns a tracked workflow with progress, ETA and a notification; the originating screen updates live	4, 5
U5	Error messages name internal objects ("Account structure does not allow this combination")	AX infolog; SAP message classes	Validation written for developers	Validation messages are metadata with user text, remediation link and the exact field; errors shown at the field as you type	3, 5
U6	Shop-floor and warehouse UIs are RF text menus or require a PC	AX `JmgRegistration` terminal; SAP ITSmobile/RF	UI frameworks predate touch and offline	Touch-first PWA kiosk and scanner apps with offline queues; one scan completes a step	25, 32
U7	Excel round-trips to do bulk edits	AX Excel add-in; SAP GUI copy/paste	No bulk-edit grid	Every list page supports multi-row inline edit, paste from spreadsheet with validation preview, and undo	5
U8	Approval requires logging in to the ERP	AX workflow work items; SAP Business Workplace (`SBWP`)	Workflow inbox is inside the client	Approvals in email/Slack/Teams/mobile with one-tap approve, full context and the same audit trail	4
U9	Copilot features are bolted on and read-only	D365 Copilot summaries; SAP Joule Q&A	AI layer has no safe action surface	Every API is an MCP tool under the user's permissions; agents draft, users approve; explain-this-record everywhere	47

B. Speed

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
S1	MRP runs overnight; planners work on stale data; a changed order waits until tomorrow	AX `ReqCalc` regeneration; SAP `MD01` total planning	Full regeneration over all items; pegging recomputed from scratch	Net-change planning engine keyed by events (order/receipt/BOM change) with incremental re-plan of affected low-level codes; full regen < 10 min for 50k items	22
S2	Inventory close/recalculation blocks posting for hours	AX `InventClosing`; SAP material ledger `CKMLCP`	Settlement of issues to receipts is a batch over all transactions	Running costing (continuous settlement on each issue for FIFO/avg) so close is a verification pass, not a computation; partitioned by item	20
S3	ATP is computed per line with locks; order entry slows under load	AX `InventOnHand`/CTP; SAP ATP with `VBBE` requirements	Row-level locking on on-hand and requirement tables	Cumulative ATP projection maintained from the inventory event stream; reads lock-free; promise computed in memory	27

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
S4	Month-end close takes 5–10 days of manual steps	AX period close checklist is a list; SAP closing cockpit sequences manual transactions	Postings are batch; reconciliations are manual	Continuous close: every subledger reconciles to the universal journal daily by machine; close checklist tasks auto-complete when their check passes	14, 16+
S5	Reports run against OLTP and time out, or against a cube that is a day old	AX SSRS on `GeneralJournalAccountEntry`; SAP BW loads	No streaming path from transaction to analytics	CDC → Iceberg lakehouse with p95 < 60 s lag; statements served from pre-aggregated balances rebuilt from the journal	11, 46
S6	Posting a 10k-line journal takes minutes and locks the period	`LedgerJournalCheckPost`; SAP `FB50` with `BSEG` inserts	Row-by-row posting through class hierarchies; synchronous number sequences	Append-only journal with batch insert, idempotency keys and asynchronous balance projection: 5,000 lines/s sustained	11
S7	Deploying a customization takes a release window	AX model store deploy with AOS restart; SAP transports with downtime	Code and metadata are compiled into the runtime	Metadata extensions apply live; code ships via GitOps with digest rollback in < 15 min	1, 3

C. Accuracy

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
A1	Subledgers drift from the GL (inventory value ≠ inventory account; AR aging ≠ AR control)	`InventSum` vs `GeneralJournalAccountEntry`; SAP `BSID` vs `SKB1` balances	Subledgers post to the GL through separate code paths with their own rounding	One posting API; subledgers are projections of the same event stream; daily machine reconciliation with zero unexplained breaks as a release gate	11, 16, 19, 30
A2	Rounding and FX differences appear as "penny" entries	AX `LedgerJournalTrans` rounding to accounting currency; SAP `BSEG` rounding differences	Rounding at line level then summing	Three-currency amounts stored per line; rounding applied once at document level with explicit rounding lines; ISO 4217 minor units enforced	12
A3	Duplicate customers/vendors/items	`CustTable` created freely; SAP `KNA1` duplicates across sales orgs	No dedup at creation; roles not layered on one party	One party per real-world entity with fuzzy-match blocking at creation (precision ≥ 0.95) and governed merge	7

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
A4	Wrong tax because tax group ↔ item tax group intersections are configured by hand	`TaxGroup`/`TaxItemGroup` intersection; SAP `FTXP` condition records	Determination is a configuration lattice with no test harness	Determination engine with a 500-scenario fixture pack reviewed by accountants; explain-this-tax on every line	13
A5	Negative inventory and phantom on-hand	`InventSum` allows negative physical when `InventModelGroup` permits; SAP `MMBE` vs `MSEG` timing	Projection updated by application code that can skip	On-hand is a projection of immutable inventory events; policy enforced at the event write; projection rebuilt from events and compared nightly	16
A6	Costing variances nobody can explain	`InventCostTrans` variances; SAP `KKS1`	Variance types computed after the fact with no linkage	Variance decomposition attached to each ended order with lot-size, quantity, price and substitution components that sum exactly	21, 39
A7	Master data changed without approval or trace	Direct edit of `VendTable` bank account; SAP `XK02`	Audit log optional (`SysDatabaseLog`) and slow	Field-level immutable audit from CDC for every entity; sensitive fields (bank accounts, credit limits) require workflow	5, 7

D. Openness and interoperability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
O1	Each interface is a project (AIF service, IDoc mapping, custom data entity)	AX AIF/`DMF` entities; SAP `WE20` partner profiles, PI/PO	Integration surface is bespoke per object	Every entity and event is published automatically (OpenAPI, AsyncAPI, MCP) from metadata; new entity ⇒ new API with no work	3, 4, 35
O2	External systems poll for changes	`DMF` change tracking; SAP change pointers (`BDCP2`)	No event stream	Kafka topics per entity with CloudEvents, replay from any offset, webhooks for partners	4
O3	MES/PLM/SCADA need a separate middleware product	SAP MII, PI/PO; AX via BizTalk/Logic Apps	ERP speaks only its own protocols	Native OPC-UA/MQTT ingestion, EDI engine, Peppol, SFTP, MCP — eight integration surfaces with one auth model	25, 35
O4	Vendor cloud lock-in (LCS, RISE)	D365 only on Microsoft-managed Azure; S/4 Cloud on RISE	Platform tied to the vendor's operations	Any conformant Kubernetes; Helm charts; self-host or managed	1

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
O5	Customizations block upgrades	X++ overlayering; ABAP modifications	Extension model is code in the vendor's layer	Extensions are metadata and sanctioned hooks; additive API guarantees; upgrades are image promotions	3
O6	Analytics requires exporting to BYOD/BW	Entity store; BW extractors	No open analytical surface	Iceberg tables readable by any engine (Trino, Spark, DuckDB, Power BI, Tableau) with governed metrics	46

0A.2 Interoperability contract ("all systems on the network can use it")

Passage treats the ERP as a service on the customer's network, not a destination. Every capability is reachable through the same eight surfaces with one identity and one policy engine:

Surface	Used by	Auth	Guarantee
REST (OpenAPI 3.1, generated per entity)	Apps, scripts, partners	OIDC bearer token or API key, OPA-authorized	p95 < 50 ms simple reads; additive versioning
GraphQL (read model)	BI tools, portals	Same	Cross-entity reads without N+1
Kafka topics (CloudEvents, AsyncAPI)	MES, WMS, data platforms, other ERPs	mTLS + ACLs per tenant	At-least-once, ordered per key, 7-day replay
Webhooks	SaaS partners	Signed (HMAC) deliveries	Retries with backoff, dead-letter, replay
MCP	AI agents (Claude and others)	Delegated user token; agent ≤ user permissions	Every tool call audited
EDI (X12/EDIFACT via AS2/SFTP) and Peppol	Trading partners, tax authorities	Partner certificates	Acknowledgments, SLA monitor
OPC-UA / MQTT	Machines, PLCs, SCADA	Device certificates	10k events/min per plant, no loss on broker restart
File / SFTP	Banks, legacy systems	Keys in Vault	ISO 20022, BAI2, NACHA, CSV with schema validation

Rule: **if it is not reachable through these surfaces, it does not ship.** Each wave's API/event section is reviewed against this rule before close.

0A.3 How "deep knowledge of AX and SAP" is applied

We deliberately keep what those systems got right and name it in every wave's parity map:

- **From AX:** the role/duty/privilege security model; inventory dimensions (product/storage/tracking) as a first-class concept; `InventTrans` status semantics; trade-agreement search order; number sequences per legal entity; financial dimensions with account structures; coverage groups; lean production flows.
- **From SAP:** the universal journal (ACDOCA) as single source of truth; parallel ledgers and currency types; condition technique for pricing and tax; posting periods per module; EWM's location-directive rigor; QM inspection lots and usage decisions; results analysis for projects; Group Reporting eliminations.

And we remove the parts that exist only because of their era: batch-first processing, table-shaped forms, code-based extensions, closed integration layers, vendor-bound operations.

Part 0 — Phase 0: Foundation & Platform Kernel (waves 1–5)

0.0 Phase overview

Business outcome at phase exit. At the end of wave 5 nothing business-facing exists, but everything a business module needs does: a change merged to `main` reaches a Kubernetes environment with digest-pinned rollback in under 15 minutes; a user signs in once through a federated IdP and lands in a legal entity; every API call is authorized by OPA against AX-style roles/duties/privileges and fenced by Postgres row-level security; an entity declared in YAML becomes a table, an API, a client and a form without hand-written code; every state change is a CloudEvent on Kafka via a transactional outbox; approvals and batch jobs run on Temporal; number sequences support gap-free mode; every field change is in an immutable, hash-chained audit log; and a click in the shell can be followed through gateway, service, database and Kafka consumer in one trace. This layer is what AX calls the AOT, SysSecurity, batch and workflow frameworks, number sequences and SysDatabaseLog, and what SAP calls DDIC, PFCG, SM36/SM37, Business Workflow, SNRO and CDHDR/CDPOS — rebuilt as separately deployable services on the ADR-0001 stack.

Phase-exit sign-off criteria

- `make deploy-local` on a clean laptop produces a working k3d environment with all Phase 0 services in under 10 minutes; the same Helm/Argo manifests are healthy in the cloud `stage` cluster.
- Stakeholder demo executes, live, the wave-5 success metric from PRD §10: deploy a change → log in via a federated IdP → extend the Vendor-placeholder entity with a field and an enum value → run a 3-level approval → open the audit entry → open the trace in Tempo.
- Table-driven authorization suite (≥ 20 duties × allow/deny) and RLS cross-tenant suite are green; effective-permission report equals OPA decision for 100 sampled resources.
- Golden-generation suite for 10 reference entities is deterministic across 3 consecutive CI runs; additive v2 change passes, breaking change fails.
- Outbox chaos test shows zero lost and zero duplicated events over 10 kill cycles; 1M-row batch resumes after pod kill with 0 duplicates.
- Gap-free number sequence Jepsen-style test (10k concurrent requests) shows 0 gaps, 0 duplicates.
- 100% of generated forms pass axe WCAG 2.1 AA; audit entry visible within 2 s (p95); every service exports OTel traces, metrics and logs.
- SLO dashboards exist for every Phase 0 service with alert routes to the PagerDuty-compatible webhook.

Cross-wave data model summary

Entity	Owning service	Key relationships
Release, Environment	platform (wave 1, GitOps repo + registry DB)	Release → Environment (promotion history)
Tenant, LegalEntity	identity (wave 2; regenerated from metadata in wave 3)	Tenant 1→N LegalEntity; every row in every domain carries tenant_id
User, Role, Duty, Privilege, RoleAssignment	identity / authz (wave 2)	User N↔N Role via RoleAssignment scoped to LegalEntity; Role 1→N Duty 1→N Privilege

Entity	Owning service	Key relationships
EntityDefinition, FieldDefinition, EnumDefinition, TypeDefinition, ExtensionPackage	metadata-registry (wave 3)	Entity 1→N Field; Field → Type or Enum; ExtensionPackage → Entity (additive)
OutboxEvent, TopicDefinition, SchemaVersion	every service / events (wave 4)	Outbox row → CloudEvent; TopicDefinition → SchemaVersion (from EntityDefinition)
WorkflowDefinition, WorkflowInstance, WorkItem, BatchJob, BatchRun, BatchTask	workflow (wave 4)	Instance → Definition version; WorkItem → Instance + assignee User; BatchRun 1→N BatchTask
NumberSequence, NumberSequenceAllocation	numseq (wave 5)	Sequence scoped to Tenant/LegalEntity; Allocation → Sequence (gap-free audit)
AuditEntry, AuditField	audit (wave 5)	Entry → any entity/record; hash chain per tenant
SavedView, Personalization, ImportJob	shell-bff / importexport (wave 5)	SavedView → EntityDefinition + User; ImportJob → EntityDefinition

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.identity.user.created.v1 / .deactivated.v1	identity	audit, workflow (assignee cache), shell	user_id, tenant_id, email, status
dev.passage.authz.role-assignment.granted.v1 / .revoked.v1	authz	audit, OPA bundle builder	user_id, role_id, legal_entity_ids, effective_from/to
dev.passage.authz.policy.compiled.v1	authz	OPA sidecars (bundle pull), audit	bundle_version, sha256, role_count
dev.passage.metadata.entity.published.v1	metadata-registry	generator, events (topic/schema), shell (form cache)	entity_name, version, diff summary
dev.passage.metadata.extension.applied.v1	metadata-registry	generator, audit, shell	package_id, entity_name, fields_added, enum_values_added
dev.passage.workflow.instance.submitted.v1 / .completed.v1	workflow	owning entity service, audit, shell inbox	instance_id, entity, record_id, outcome
dev.passage.workflow.workitem.assigned.v1 / .escalated.v1	workflow	shell inbox, notifications	workitem_id, assignee_id, due_at
dev.passage.batch.run.completed.v1 / .failed.v1	workflow (batch)	shell job monitor, alerting	job_id, run_id, rows_processed, error
dev.passage.audit.entry.recorded.v1	audit	shell audit viewer (live), analytics (wave 46)	entry_id, entity, record_id, operation, hash

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Metadata kernel under-specified before business waves need it	Medium	High	10 reference entities in wave 3 include the shapes of Party, Product, JournalLine; golden tests frozen at phase exit	Product
Two-person team cannot operate Kafka + Temporal + Keycloak + CNPG	Medium	High	All operators installed via Helm umbrella; runbooks per component; k3d parity so every incident reproduces locally	Engineering
OPA policy compile from roles produces unexpected denies	Medium	Medium	Effective-permission viewer and `decision` debug endpoint; deny-reason strings in every 403	Engineering

Risk	Likelihood	Impact	Mitigation	Owner
RLS misconfiguration leaks tenant data	Low	Critical	`FORCE ROW LEVEL SECURITY` on every table; app role NOBYPASSRLS; CI test queries every table cross-tenant	Security
Gap-free sequence becomes a throughput bottleneck	Medium	Medium	Continuous mode only where fiscal law requires it; non-continuous is the default with block pre-allocation	Engineering
Audit volume from CDC overwhelms Postgres	Medium	Medium	Monthly partitions; audit flag per field; PII masking at capture; 10-year retention tiered to object storage from wave 46	Engineering

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0.1 Wave 1: Delivery Pipeline & Cluster Baseline

0.1.1 Scope and business context

AX ships code as LCS deployable packages built on a build VM and applied with AxUpdateInstaller; model store exports (`AxUtil export`) and database restores are the rollback. SAP moves ABAP through STMS transport requests (SE09/SE10 workbench requests, DEV → QAS → PRD routes, `tp import`), with back-out by transport of copies. Both couple release to a vendor tool and give no software bill of materials, no cryptographic signature and no sub-hour rollback. Wave 1 replaces this with a GitOps pipeline on any conformant Kubernetes. At the end of the wave an engineer can merge a pull request and watch the change reach `dev`, promote the same image digest to `stage`, and roll back to the previous digest, each in under 15 minutes. Out of scope: multi-region, production-grade HA Postgres, secrets rotation policy, cost reporting, and any business service other than the `hello-ledger` placeholder.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Applying an LCS deployable package to a D365 sandbox takes 2–4 h of downtime and a service request for production; a failed apply means a database restore	Package = whole-system binary; environment managed by vendor tooling	Per-service OCI image + Helm chart, promoted by digest through Argo CD; rolling update with no downtime	Merge → running in env: 4 h → < 15 min; downtime per release: hours → 0
STMS transport import to PRD (STMS_IMPORT) is a manual queue with no automatic back-out; a bad transport is reversed by hand-building a transport of copies	Transports carry object deltas, not immutable artifacts	Immutable digests; rollback is a one-line overlay revert	Rollback: 1–3 h and a Basis ticket → < 5 min, self-service
No SBOM or signature for an AX model or SAP transport; security teams cannot answer "which CVE is in production"	Build tooling predates supply-chain attestation	Syft SBOM, Grype/Trivy gate, cosign signature verified at admission	CVE-to-affected-environment lookup: days of manual inventory → one query; unsigned artifacts in prod: 0
OneBox/VHD developer environments (AX) or NetWeaver dev editions (SAP) take days to provision and drift from production	Monolithic app server image per developer	`make deploy-local` on k3d with the same charts as cloud	New engineer to a running environment: 2–5 days → < 10 min

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Customers cannot host on their own infrastructure without LCS (D365) or RISE (S/4)	Vendor-managed cloud is the only supported topology	One OpenTofu interface across k3d/EKS/GKE/AKS/bare-metal; nothing provider-specific in the critical path	Supported deployment targets: 1 → 5 with identical manifests

0.1.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Package a release	LCS deployable package, `AxUtil export` model store, AXModel files	STMS transport request (E070/E071), `tp export`	OCI image (digest) + Helm chart version in `deploy/`	improved
Build and validate	Build VM, `AxBuild.exe`, Best Practice checks	SE80 activation, ATC (code inspector SCI), ABAP unit	GitHub Actions: lint, unit, container build, SBOM, scan	improved
Environment tiers	LCS dev/test/UAT/prod topology	STMS transport route DEV→QAS→PRD, STMS_QA	Argo CD app-of-apps per environment: dev, stage, prod	full
Promote between tiers	LCS "apply package" to the next environment	STMS import queue, `tp import`, STMS_IMPORT	`passage promote --to stage --digest sha256:…` opens a PR against the env overlay	full
Rollback	Restore DB + model store, "AOT revert"	Transport of copies back-out, SNOTE de-implement	Argo CD rollback to prior digest, `passage rollback --env stage`	improved
Code signing / trust	Strong-name model signing (`AxUtil … /key`)	Digitally signed SAP Notes, TMS trusted systems	cosign keyless signatures, policy-controller admission denies unsigned images	improved
Hotfix path	KB hotfix via LCS, binary update	SNOTE note implementation, SPAU	Hotfix branch → cherry-pick → same pipeline, same gates	full
Deployment log / history	LCS environment history, AxUpdateInstaller logs	STMS import log, SLOG/ALOG	Argo CD history + GitHub Deployments API + `platform.release` table	improved
Local dev environment	OneBox VM / VHD from LCS	SAP NetWeaver developer edition, mini-basis	`make deploy-local` on k3d, same charts as cloud	improved

0.1.3 Functional requirements

- **FR-1.1 Monorepo layout** — Must. The system shall keep all Passage source in one repository with the layout `services/go/<svc>`, `services/ts/<svc>`, `services/py/<svc>`, `web/shell`, `web/remotes/<name>`, `deploy/charts/<svc>`, `deploy/envs/{dev,stage,prod}`, `infra/modules/<provider>`, `metadata/entities`, `docs/`.
  - Notes: each service owns a `Dockerfile`, `Makefile`, `CODEOWNERS` entry and a Helm chart; CI path filters build only changed services; a `go.work` and `pnpm` workspace root pin toolchain versions (Go 1.23, Node 22, Python 3.12).
- **FR-1.2 Cluster bootstrap module** — Must. The system shall provide an OpenTofu module `infra/modules/cluster` with one variable interface (`provider ∈ {k3d, eks, gke, aks, baremetal}`, `name`, `node_pools[]`, `kubernetes_version`) that outputs a kubeconfig and installs the base add-ons.
  - Notes: base add-ons are Argo CD, Envoy Gateway, cert-manager, External Secrets Operator, CloudNativePG operator, sigstore policy-controller, metrics-server; provider modules contain only networking, node pools and storage classes; k3d module runs with 3 agent nodes and 8 GB RAM.
- **FR-1.3 Argo CD app-of-apps** — Must. The system shall define a root Application per environment that reconciles child Applications from `deploy/envs/<env>/apps/*.yaml`, each pinning a chart version and an

image digest.
    - Notes: sync policy automated for dev, manual-approve PR for stage and prod; prune and self-heal enabled; sync waves order operators (–2), databases (–1), services (0), gateway routes (1).
- **FR-1.4 Digest-only promotion** — Must. The system shall promote images by immutable digest; tags are informational only.
    - Notes: `passage promote` rewrites `image.digest` in the target overlay and opens a PR with the SBOM diff and scan summary; promotion to prod requires the digest to have run in stage for ≥ 60 minutes with no SLO burn alert.
- **FR-1.5 CI pipeline** — Must. The system shall run on every PR and on `main`: lint (golangci-lint, eslint, ruff), unit tests with coverage ≥ 70% per service, container build (BuildKit, reproducible), Syft SBOM (SPDX 2.3), Grype and Trivy scan, cosign sign and attest.
    - Notes: required checks block merge; `main` builds push to the registry under `ghcr.io/passage/<svc>@sha256:…`; median PR pipeline ≤ 8 minutes; cache keyed by lockfile hash.
- **FR-1.6 Vulnerability gate** — Must. The system shall fail the build on any CRITICAL finding, or any HIGH finding older than 14 days wit